

3.1.6 Ecosystems under stress

This optional section of our specification focuses on the biosphere and in particular the nature and functioning of ecosystems and their relationships to the nature and intensity of human activities. Study of the impact of population growth and economic development on ecosystems at various scales affords the opportunity for students to engage with fundamental contemporary people–environment issues including those relating to biodiversity and sustainability. Study of this section offers the opportunity to exercise and develop observation skills, measurement and geospatial mapping skills, together with data manipulation and statistical skills including those associated with and arising from fieldwork.

3.1.6.1 Ecosystems and sustainability

The concept of biodiversity. Local and global trends in biodiversity. Causes, rates and potential impacts of declining biodiversity.

Ecosystems and their importance for human populations in the light of continuing population growth and economic development. Human populations in ecosystem development and sustainability.

3.1.6.2 Ecosystems and processes

Nature of ecosystems – their structure, energy flows, trophic levels, food chains and food webs.

Application of systems concepts to ecosystems – inputs, outputs, stores and transfers of energy and materials. Concepts of biomass and net primary production.

Concepts of succession: seral stages, climatic climax, sub-climax and plagioclimax.

Mineral nutrient cycling.

Nature of terrestrial ecosystems and the inter-connections between climate, vegetation, soil and topography which produce them. Ecosystem responses to changes in one or more of their components or environmental controls.

Factors influencing the changing of ecosystems, including climate change and human exploitation of the global environment.

3.1.6.3 Biomes

The concept of the biome. The global distribution of major terrestrial biomes.

The nature of two contrasting biomes: tropical rainforest and savanna grassland to include:

- the main characteristics of each biome
- ecological responses to the climate, soil and soil moisture budget – adaptations by flora and fauna
- human activity and its impact on each biome
- typical development issues in each biome to include changes in population, economic development, agricultural extension and intensification, implications for biodiversity and sustainability.

3.1.6.4 Ecosystems in the British Isles over time

Succession and climatic climax as illustrated by lithoserres and hydrosere.

The characteristics of the climatic climax: temperate deciduous woodland biome.

The effects of human activity on succession – illustrated by one plagioclimax such as a heather moorland.

3.1.6.5 Marine ecosystems

The distribution and main characteristics of coral reef ecosystems. Environmental conditions associated with reef development.

The following aspects should be examined with reference to a named, located coral reef:

Factors in the health and survival of reefs:

- Natural: Water temperature, acidity, salinity, algal blooms.
- Human activity and its impact: Major drainage basin schemes, onshore development, desalination, pollution, tourism, fishing.
- Future prospects for coral reefs.

3.1.6.6 Local ecosystems

The main characteristics of a distinctive local ecosystem (such as an area of heathland, managed parkland, pond, dune system). Ecological responses to the climate, soil and soil moisture budget – adaptations by flora and fauna.

Local factors in ecological development and change (such as agriculture, urban change, the planned and unplanned introduction of new species).

The impacts of change and measures to manage these impacts. Conservation strategies and their implementation in specific settings.

3.1.6.7 Case studies

Case study of a specified region experiencing ecological change to illustrate and analyse the nature of the change and the reasons for it, how the economic, social and political character of its community reflects its ecological setting and how the community is responding to change.

Case study of a specified ecosystem at a local scale to illustrate and analyse key themes set out above, including the nature and properties of the ecosystem, human impact upon it and the challenges and opportunities presented in its sustainable development.